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Mid Semester Examination March 2024

Name of the Program: B.Tech B.Tech (Biotech) Semester: VI
Name of the Course: Statistical Data Analysis with R Course Code: TCS-471

Time: 1 Hour 30 Minutes Maximum Marks: 50

Note:
(i) Answer **all the questions** by choosing ***any one of the sub questions***.
(ii) Each question carries 10 marks

Q1	(10 marks)	CO1
a)	Compare and contrast descriptive and summary statistics..	
	OR	
b)	Why statistical analysis is required in digital era ? State the concept of Uni-variate, bi-variate sampling and confidence interval.	
Q2	(10 marks)	CO3
a)	The number of ATM transactions per day were recorded at 15 locations in a large city. The data were: 35, 45, 125, 48, 35, 65, 43, 55, 52, 76, 48, 425, 470, 32, and 60. Find (a) the median number of transactions and (b) the mean number of transactions	
	OR	
b)	What is re-sampling? Discuss the various methods of Re-sampling ?	
Q3	(10 marks)	CO3
a)	A and B play 12 games of chess, of which 6 are won by A, 4 are won by B, and 2 end in a draw. They agree to play a match consisting of 3 games. Find the probability that (a) A wins all 3 games, (b) 2 games end in a draw, (c) A and B win alternately, and (d) B wins atleast 1 game	
	OR	
b)	A special dice is prepared such that the probabilities of throwing 1, 2, 3, 4, 5 and 6 points are: (1-k)/6, (1+2k)/6, (1-k)/6, (1+k)/6, (1-2k)/6 and (1+k)/6 respectively. If two such dice are thrown, find the probability of getting a sum equal to 9.	
Q4	(10 marks)	CO2
a)	Which factors contribute to the preference for R as a language for statistical analysis and visualization over Python?	
	OR	
b)	Compare standard packages and command packages in R ? Elaborate the steps for Installing Extra Packages for Windows Users.	
Q5	(10 marks)	CO3
a)	Specify the use of c (), scan(), read.csv() and read.table() in R with proper syntax for numeric and character data with suitable examples.	
	OR	
b)	You have the results of a simple experiment to look at the visitation of various vehicles to different parking lot. The number of vehicles observed was as follows: ISBT: 10 1 37 5 12 TERMINAL T1: 8 3 19 6 4 TERMINAL T2: 18 9 1 2 4 TERMINAL T3: 12 13 16 9 10 TERMINAL T4: 8 27 6 32 23 Write the syntax in R to: **A.** Creation of five simple numeric vectors of these data. **B.** Save all the items you just created to a disk file in your working directory. **C.** To remove all the vectors that you just made. **D.** To recall the vectors from disk.	